

ORIGINAL ARTICLE

Statins Use in Veterans With Peripheral Artery Disease: Insights From the PEARLS Study

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BACKGROUND: Guidelines recommend high-intensity statins in patients with peripheral artery disease (PAD). We examined contemporary use of statins among real-world patients with a new diagnosis of PAD.

METHODS: Within a national registry of Veterans with incident PAD, we used pharmacy claims data to determine active statin use within 1 year of PAD diagnosis. We also used laboratory data to evaluate lipid levels. Hierarchical multivariable logistic regression was used to identify variables associated with the use of any statin and high-intensity statin 1 year after PAD diagnosis and quantify variation in statin use across sites.

RESULTS: Among 91 970 patients with PAD alive at 1 year, the median age was 70 years, 97.2% were men, and 18.6% were Black. At 1 year, 64 231 (69.9%) patients were on a statin, and 34 017 (37.0%) were on a high-intensity statin. Among nonstatin users, 53.4% (14 834/27 739) had not been prescribed a statin at any time in the preceding 3 years. Overall, the median low-density lipoprotein cholesterol was 81.6 mg/dL and was lower in those on high-intensity statins versus low/moderate-intensity statins versus no statins (72.2 versus 78.0 versus 100 mg/dL). Age, prior history of coronary artery disease, cerebrovascular disease, hypertension, diabetes, and current or former smoking history, obesity, and revascularization during the initial 6 months were associated with higher statin and high-intensity statin use (among statin users). At the site level, there was modest variation in statin use, especially high-intensity statins, which ranged from 20% to 54% with an overall median odds ratio of 1.39 (95% CI, 1.32–1.46).

CONCLUSIONS: Nearly 70% of patients with PAD were receiving a statin within 1 year of PAD diagnosis. However, use of high-intensity statins and nonstatin lipid-lowering therapy remained suboptimal. Moreover, there was modest variation in use of statins, especially high-intensity statins, across sites, highlighting opportunities for improving guideline-concordant care in patients with PAD.

Key Words: aged ■ cholesterol ■ peripheral artery disease ■ statins ■ Veterans

Randomized controlled trials show that high-intensity statins reduce the risk of clinical events in patients with atherosclerotic cardiovascular disease (ASCVD).^{1,2} Patients with peripheral artery disease (PAD) represent a high-risk ASCVD subgroup with increased cardiovascular and limb event risk,^{3–6} which is reduced by high-intensity statin therapy.^{7–10} Accordingly, PAD guidelines give high-intensity statins a Class I recommendation for patients with PAD.^{11,12}

Despite the strong guideline recommendations, studies have found a low rate of statin use in PAD.^{7,10,13,14} In a recent administrative claims data study, only 46% of patients with PAD were on a statin, and only 23.3% were on a high-intensity statin.¹³ Although these data have been used to highlight deficiencies in risk-factor management in patients with PAD, the extent to which low statin use in these studies was due to the inclusion of patients with non-PAD who may not have an indication

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WHAT IS KNOWN

- Current guidelines strongly recommend high-intensity statins (Class I) in patients with peripheral artery disease.
- Although prior studies report low rates of statin use, these studies have been limited by the use of diagnostic codes that have limited accuracy for identifying patients with peripheral artery disease.

WHAT THE STUDY ADDS

- Among 91 970 Veterans with a new diagnosis of peripheral artery disease identified using a highly accurate natural language processing system, we found that ≈70% of the patients were actively receiving a statin 1 year after peripheral artery disease diagnosis.
- However, rates of high-intensity statin and nonstatin lipid-lowering therapy use were low, and fewer than 40% achieved low-density lipoprotein cholesterol <70 mg/dL.
- There was modest variation in the use of high-intensity statins across sites even after adjustment for patient variables, highlighting the importance of facility-level factors in site-level variation in statin use.

Nonstandard Abbreviations and Acronyms

ABI	ankle-brachial index
ASCVD	atherosclerotic cardiovascular disease
BMI	body mass index
CAD	coronary artery disease
CeVD	cerebrovascular disease
LDL	low-density lipoprotein
PAD	peripheral artery disease
PEARLS	Peripheral Artery Disease Long-term Survival
TBI	toe-brachial index
VA	Veterans Affairs
VHA	Veterans Health Administration

for statins remains unclear. This is because PAD was identified using diagnostic codes that have limited accuracy, with positive predictive values ranging from 27% to 76%.^{15–18} Moreover, statin use was assessed based on prescriptions in electronic health record systems, which may not reflect actual use. Given the vital role of statins in the management of PAD, a comprehensive understanding of statin use in patients with PAD is warranted. To address this knowledge gap, we leveraged the PEARLS study (Peripheral Artery Disease Long-Term Survival) data set, a large registry of Veterans with new-onset PAD identified using abnormal ankle-brachial

index (ABI) and toe-brachial index (TBI) values extracted using a validated natural language processing system (positive predictive value 92.3% for PAD identification) at the Veterans Health Administration (VHA).¹⁹ The VHA is the largest integrated healthcare system in the United States, comprising >9 million enrollees, of whom 6 million actively receive care in Veterans Affairs (VA). This represents roughly 2% of the overall US population.^{20–22} The PEARLS registry includes rich data on clinical variables, including longitudinal prescription fill data from the VHA and Medicare Part D, which we utilized to: (1) define patterns of statin and high-intensity statin use in patients newly diagnosed with PAD at 1 year following diagnosis, (2) determine patient characteristics associated with statin use, and (3) describe site-level variation in statin utilization.

METHODS

This work was approved by the institutional review board at UT Southwestern Medical Center and Research & Development Committees at Dallas, Iowa City, and Tennessee Valley Health System VA Hospitals. We followed the Strengthening the Reporting of Observational Studies in Epidemiology observational checklist when reporting our findings, as presented in Table S1. Because of the sensitive nature of the data collected for this study and the VHA stipulation regarding disclosure of the data set, we are unable to make the study data publicly available. All requests to access the study data set must be directly sent to the VHA at vinci@va.gov.

Data Sources

We obtained the relevant clinical information for patients in our cohort through the VA corporate data warehouse, which contains information on inpatient and outpatient encounters, pharmacy prescription fills, and laboratory results. Since Veterans often receive care at non-VA facilities, we also obtained data from Fee-basis files and consolidated data sets files for clinical encounters at non-VA sites paid by the VA and Medicare (Parts A–D in those with Medicare coverage).

Cohort Development

The details of the PEARLS registry have been described previously.²³ Briefly, we identified 103 478 patients with a new diagnosis of PAD from 2015 to 2020 using a validated natural language processing system that extracts ABI and TBI values from test reports. An ABI value of ≤0.9 or a TBI value of ≤0.7 in any limb is used to define PAD.¹⁹ The date of the first abnormal ABI or TBI consistent with a diagnosis of PAD was defined as the index date. Since clinical recognition of PAD following an abnormal ABI test may lag, we evaluated statin use 1 year after the index date (ie, date of PAD diagnosis). Accordingly, only patients alive 1 year after PAD diagnosis were included. We also restricted our cohort to patients with at least 1 outpatient clinic visit (primary care, specialty medicine, or surgery) and who had filled at least 1 prescription for any medication (not just statins) within 1 year after PAD diagnosis, to ensure patients were receiving follow-up healthcare within the VA.

Finally, we also excluded VA facilities (and patients from those facilities) with fewer than 100 patients over a 6-year study period. Our final cohort included 91 970 patients from 115 VA sites (Figure 1).

Study Variables and Outcomes

The primary outcome of the study was statin use 1 year after the index date. We chose the 1-year time frame to allow treating physicians to have the opportunity to initiate statins in nonusers following diagnosis of PAD. We used VA Pharmacy and Medicare Part D claims data to determine statin use. Both data sources include information on drug classes, names, doses, supply quantities/d, and prescription fill dates. We classified a patient as an active statin user 1 year after the index date if they had filled a statin prescription within 2× the number of days supplied by that prescription before that 1-year mark. For example, a patient would be considered an active statin user at the 1-year post-index date if they had filled a 90-day statin prescription within 180 days before the 1-year mark from the index date. We used the same definition to identify active statin users at the index date. High-intensity statin was defined as atorvastatin ≥40 mg, rosuvastatin ≥20 mg, or simvastatin ≥80 mg, while all other statins or doses were considered low/moderate-intensity statins. We also obtained data on statin intolerance (history of allergy or adverse effect) up to the index PAD diagnosis documented in the allergy tables in the corporate data warehouse. Finally, we also evaluated the use of non-statin lipid-lowering therapy, including ezetimibe and PCSK-9 (proprotein convertase subtilisin/kexin type 9) inhibitors.

Study variables included patient demographics (age, sex, race, and ethnicity), smoking status, body mass index (BMI), and cholesterol levels. All variables were assessed at 1 year post-index PAD diagnosis to align with the assessment of statin use. We also assessed the presence of clinical comorbidities, coronary artery disease (CAD), cerebrovascular disease (CeVD), diabetes, hypertension, chronic kidney disease, chronic obstructive pulmonary disease, heart failure, cancer, dementia, depression, and chronic liver disease. We identified these comorbidities using the Elixhauser and Quan algorithms for claims data^{24,25} and *International Classification of Diseases* diagnosis codes from inpatient and outpatient encounters during a 2-year period before the time when statin use was assessed (1 year post-index PAD diagnosis). We also updated the definition of CAD to include a prior history of percutaneous coronary intervention and coronary artery bypass graft, which were identified using *International Classification of Diseases* procedure codes and current procedural terminology codes. Lower-extremity revascularization procedures during the first 6 months after index PAD diagnosis were identified using appropriate *International Classification of Diseases* procedure codes and current procedural terminology codes. The 6-month window was chosen to ensure that the assessment of revascularization (covariate) preceded the outcome (ie, statin group). Since all patients in our cohort had newly diagnosed PAD, ASCVD was based on the additional presence of CAD and CeVD. Smoking status was determined using the health factors table in the corporate data warehouse data. BMI values represent the most recent data recorded in the corporate data warehouse before the 1-year post-index date. We used laboratory data for LDL (low-density lipoprotein), HDL (high-density

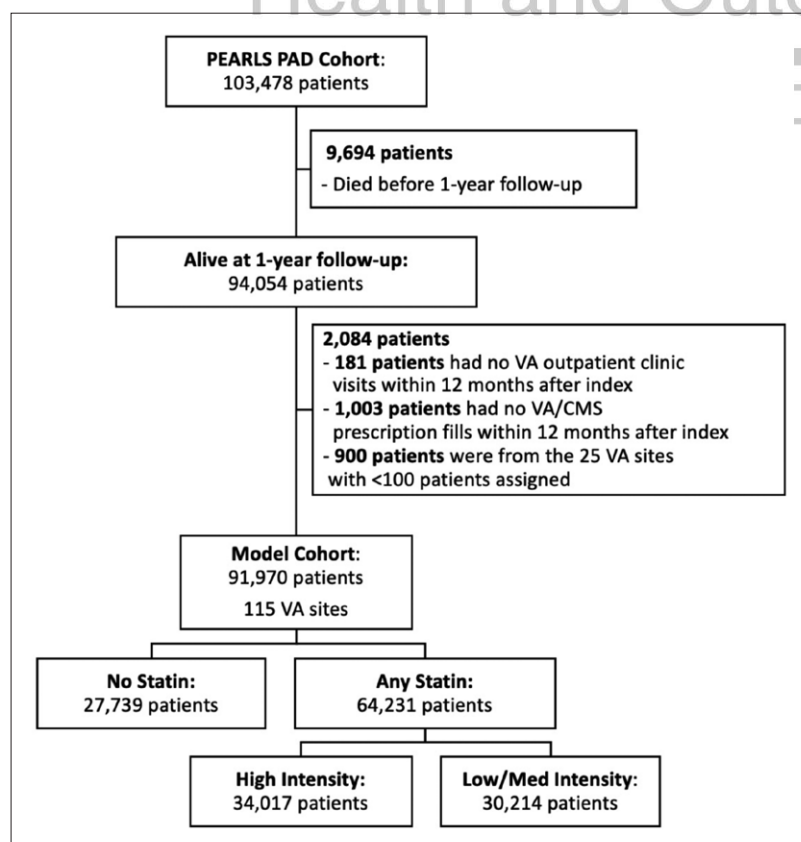


Figure 1. Development of the study cohort.

Among patients in the PEARLS registry (Peripheral Artery Disease Long-Term Survival), we excluded patients who died within 1 year of index peripheral artery disease (PAD) diagnosis, those without regular follow-up at the Veterans Affairs (VA), and those without at least 1 VA or Medicare Part D pharmacy fill. CMS indicates Centers for Medicare & Medicaid Services.

lipoprotein), and total cholesterol levels at baseline and at 1 year following the index PAD diagnosis. Severity of PAD was defined based on index ABI and TBI values in the worst limb as follows: severe: $0 \leq \text{ABI} < 0.5$ or $0 \leq \text{TBI} < 0.4$; moderate: $0.5 \leq \text{ABI} < 0.8$ or $0.4 \leq \text{TBI} < 0.6$; and mild: $0.8 \leq \text{ABI} \leq 0.9$ or $0.6 \leq \text{TBI} \leq 0.7$. Finally, we assigned each patient to a VA facility based on the site where they received the majority of their care between the index date for PAD and the 1-year follow-up.

Statistical Analysis

First, we used descriptive statistics to describe patient characteristics stratified by statin use (high-intensity, low/moderate-intensity, and no statin) at 1 year after PAD diagnosis. We used medians (interquartile range) for continuous variables and n (%) for categorical variables. We also examined the longitudinal trajectory of statin and high-intensity statin therapy use from PAD diagnosis (index date) to 1 year along with the corresponding LDL cholesterol levels across different statin trajectories. Among patients who were not on statins at 1 year after PAD diagnosis, we accounted for the possibility that statins were not prescribed (or had been previously discontinued) due to side effects by evaluating the proportion of these patients who had ever been on a statin during the preceding 3 years. We also evaluated the proportion of patients with a previously documented allergy or adverse effects from prior statin use.

We also compared the use of nonstatin lipid-lowering therapy and LDL, HDL, and total cholesterol levels among no statin, low/moderate-intensity statin, and high-intensity statin groups. Further, we evaluated differences in the statin utilization rates and cholesterol levels stratified by the presence of concomitant ASCVDs. We used the Kruskal-Wallis test to compare continuous variables and χ^2 test to compare categorical variables across these subgroups.

Next, to evaluate patient variables associated with statin use and high-intensity statin use, we constructed separate 2-level multivariable hierarchical logistic regression models. The model for any statin use included all patients, while the model for high-intensity statin use included only patients receiving a statin. To account for clustering of patients, we included VA facility as a random effect and the following patient factors as fixed effects: demographics (age, sex, race, ethnicity), comorbidities (CAD, CeVD, diabetes, hypertension, chronic kidney disease, chronic obstructive pulmonary disease, heart failure, cancer, dementia, depression, and liver disease), smoking status, severity of PAD, BMI, and lower extremity revascularization within 6 months of PAD diagnosis. Estimates from these models were reported as odds ratios with 95% CIs, and $P < 0.05$ was considered statistically significant.

We also assessed site-level variation in statin use by examining the proportion of patients with PAD actively using any statin, and high-intensity statin, 1 year following PAD diagnosis. For this analysis, patients were allocated to the VA site as described above. We reported median facility-level statin utilization rates and their ranges. To quantify the magnitude of facility-level variation after accounting for between-facility differences in patient characteristics, we calculated the median odds ratio using the estimate of the facility random effect from the aforementioned hierarchical model (for both any statin and high-intensity statin use).^{26,27} The median odds ratio is always ≥ 1 ; a median odds ratio of 1 indicates no site-level variation.

The median odds ratio can be conceptualized as the relative odds that a patient from a randomly selected facility receives a statin, compared with another patient with the same covariates from another randomly selected facility. For example, a median odds ratio of 1.50 suggests that the odds of receiving statins vary by 50% between facilities for the same patient. The median odds ratio overcomes the interpretational limitations of the intraclass correlation coefficient for categorical data. Also, it permits meaningful comparisons of hospital variation with the effect size of patient variables (eg, sex) included in the hierarchical models.^{27,28}

To further assess significant between-hospital variation in statin use and identify the number of hospitals that performed significantly better or worse compared with the national average, we estimated each hospital's adjusted rate of statin use as the ratio of the predicted-to-expected number of statin users multiplied by the overall unadjusted rate of statin use. The predicted number of statin users was estimated using the patient characteristics at the hospital and its random intercept, whereas the expected number of statin users was estimated using patient characteristics and the average hospital intercept under the assumption that the patients at that hospital were treated at the average hospital. Next, we determined the 95% CI for each facility's predicted-to-expected ratio using a nonparametric bootstrap with 500 iterations. Facilities were classified as significantly different if the 95% CI around the facility's predicted-to-expected ratio did not include 1 (ie, either higher or lower), indicating deviation from the national average. All analyses were done using SAS, version 9.4 (SAS Institute, Cary, NC) and R, version 4.4.0.

RESULTS

A total of 91 970 patients with a new diagnosis of PAD were alive and had at least 1 health care contact and pharmacy refill over 1 year following a new PAD diagnosis. Table 1 describes the characteristics of our study cohort. The median age was 70 years. The majority of the patients were male (97.2%) and self-identified as White (74.5%). There was a high prevalence of hypertension (90.5%), diabetes (57.4%), CAD (49.2%), CeVD (30.1%), chronic kidney disease (27.7%), and a current or former (60.8%) history of smoking. 10 206 (11.2%) patients in our cohort underwent lower extremity revascularization within 6 months of PAD diagnosis. The distribution of these characteristics stratified by statin use (high-intensity, low/moderate-intensity, and no statin) is presented in Table S2.

Figure 2 shows the trajectory of statin use from initial PAD diagnosis over a 1-year follow-up. At the index date, 61 297 (66.6%) were active statin users (high-intensity, 29 270 [31.8%]). At 1 year, 9768 (31.8%) of nonusers of statins at the index date became newly active, while 4077 (12.7%) of patients on low/moderate-intensity statins intensified statin therapy. However, 6834 (11.1%) of statin users at index were no longer active at 1 year. At 1 year, statin therapy remained unchanged in 86.1% of high-intensity statin users and

Table 1. Characteristics of Patients 1-Year Following PAD Diagnosis

Characteristic	Patient characteristics (N=91 970)
Age, y; median (IQR)	70 (10)
<55	2349 (2.6%)
55–64	16 434 (17.9%)
65–74	46 150 (50.2%)
≥75	27 037 (29.4%)
Sex, n (%)	
Female	2556 (2.8)
Male	89 414 (97.2)
Race, n (%)	
Black	17 141 (18.6)
White	68 521 (74.5)
Other or unknown	6308 (6.9)
Ethnicity, n (%)	
Hispanic or Latino	2740 (3.0)
Not Hispanic or Latino	85 994 (93.5)
Unknown	3236 (3.5)
ASCVD subtype, n (%)	
No prior ASCVD	35 988 (39.1)
Prior CAD only	28 298 (30.8)
Prior CeVD only	10 709 (11.6)
Prior CAD and CeVD	16 975 (18.5)
Comorbidities, n (%)	
Diabetes	52 780 (57.4)
Hypertension	83 247 (90.5)
Chronic kidney disease	25 467 (27.7)
COPD	38 137 (41.5)
Heart failure	24 694 (26.9)
Cancer	17 242 (18.7)
Depression	32 917 (35.8)
Dementia	5762 (6.3)
Liver disease	10 896 (11.8)
Smoking history, n (%)	
Never	14 526 (15.8)
Current	21 742 (23.6)
Former	34 251 (37.2)
Unknown	21 451 (23.3)
Severity of PAD*, n (%)	
Mild	27 555 (30.0)
Moderate	44 482 (48.4)
Severe	19 933 (21.7)
Revascularization within 6 months of an index PAD diagnosis, n (%)	10 286 (11.2)
Body mass index	
Median (IQR)	28.4 (7.7)
Unknown BMI, n (%)	6224 (6.8)
BMI ≥30, n (%)	33 686 (36.6)

(Continued)

Table 1. Continued

Characteristic	Patient characteristics (N=91 970)
Lipid values, median (IQR)	
LDL-C, mg/dL	81.6 (45.0)
HDL-C, mg/dL	40.7 (16.0)
Total cholesterol, mg/dL	152.0 (55.0)
LDL >70 mg/dL, n (%)	57 565 (62.6%)

SI conversion factors: to convert cholesterol levels to mmol/L, multiply values by 0.0259. ABI indicates ankle-brachial index; ASCVD, atherosclerotic cardiovascular disease; BMI, body mass index; CAD, coronary artery disease; CeVD, cerebrovascular disease; COPD, chronic obstructive pulmonary disease; HDL-C, high-density lipoprotein cholesterol; IQR, interquartile range; LDL-C, low-density lipoprotein cholesterol; PAD, peripheral artery disease; and TBI, toe-brachial index.

*PAD Severity is defined using ABI and TBI values in the worse limb at the time of PAD diagnosis as follows: severe: $0 \leq \text{ABI} < 0.5$ or $0 \leq \text{TBI} < 0.4$; moderate: $0.5 \leq \text{ABI} < 0.8$ or $0.4 \leq \text{TBI} < 0.6$ and; mild: $0.8 \leq \text{ABI} \leq 0.9$ or $0.6 \leq \text{TBI} \leq 0.7$.

75.6% of low/moderate-intensity statin users at index, while 68.2% of nonusers of statins remained nonusers. Overall, at 1 year, 64 231 (69.8%) patients were active statin users (high-intensity, 34 017 [37.0%]). The majority of statin prescriptions (91.7%) at 1 year after index PAD diagnosis were filled through the VA, with the remaining 8.3% filled via Medicare Part D. Among 27 739 nonstatin users at 1 year, 53.4% had not received a statin at any time during the previous 3 years. Overall, 16.2% of statin nonusers and 12.6% of active users had a documented allergy or adverse event due to statins (Table S3). After excluding patients with a documented allergy or adverse effect to a statin, we found that overall statin use was 70.7% and high-intensity statin use was 38.1% (Table S4).

The median LDL-cholesterol in our cohort was 81.6 mg/dL (to convert to mmol/L, multiply by 0.0259) and was above the guideline-recommended 70 mg/dL in 63.7% of patients. The median LDL-cholesterol was lower in those on high-intensity statins compared with those in the low/moderate-intensity statin and no statin groups (72.2 mg/dL versus 78.0 mg/dL versus 100.0 mg/dL; $P < 0.01$; Table S5). As expected, the proportion of patients not achieving the guideline-directed therapeutic LDL targets of ≤ 70 mg/dL (or ≤ 55 mg/dL) was lower in the high-intensity statin group compared with the low/moderate-intensity or no statin groups. LDL changes from baseline to 1 year following index PAD diagnosis followed the expected course when stratified by statin trajectories (Table S6). Overall, the use of ezetimibe and PCSK-9 inhibitors in our cohort was relatively low at 1.8% and 0.0%, respectively. When stratified by the presence of concomitant ASCVD, statin use was highest in patients with both CAD and CeVD, followed by those with CAD only and CeVD only, while patients with a new diagnosis of PAD in the absence of preexisting ASCVD had the lowest statin use (80.3% versus 77.4% versus 69.8% versus 59.0%; $P < 0.01$; Table S7). Similar trends were seen in high-intensity statin use and the proportion

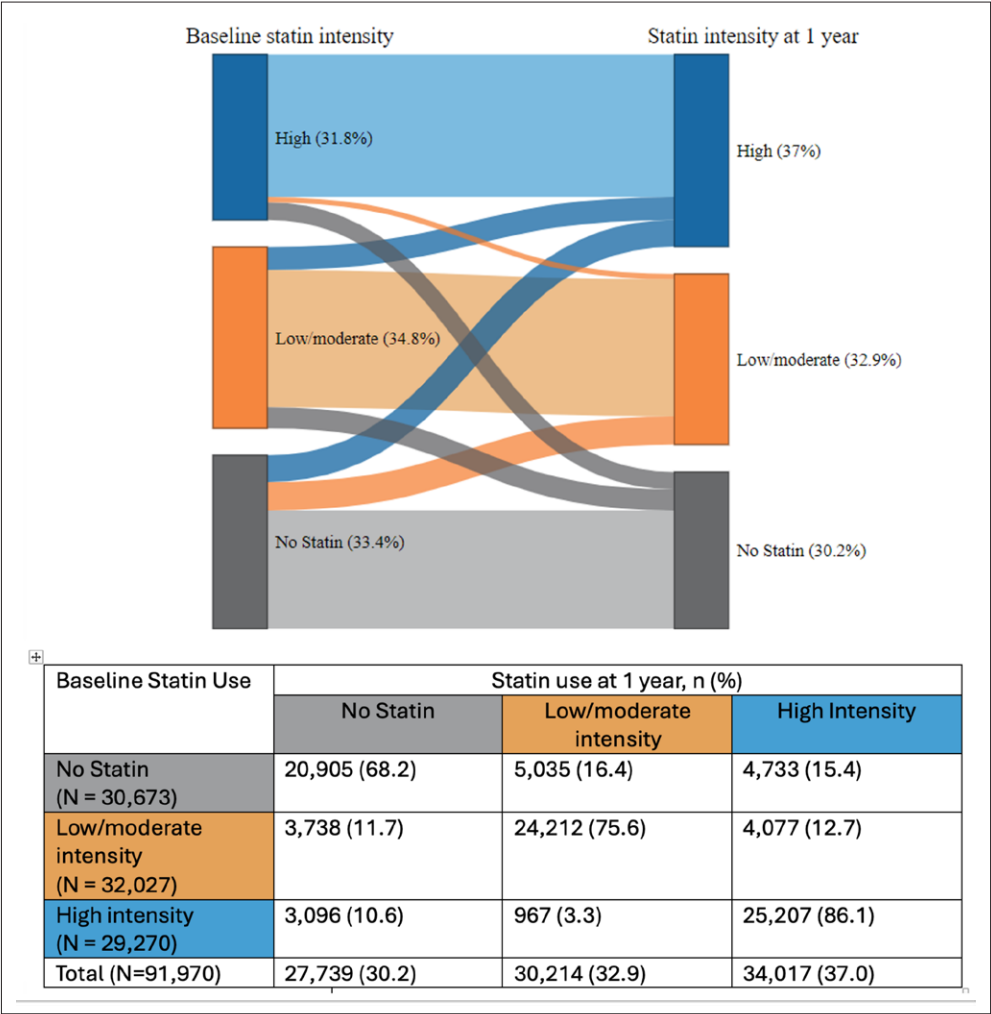


Figure 2. Trajectory of changes in statin intensity over 1 year since an index peripheral artery disease (PAD) diagnosis.

The majority of patients did not have any changes in their statin therapy within 1 year of an index PAD diagnosis. Only a minor proportion of patients had statin therapy initiated or intensified.

of patients achieving goal-directed therapeutic cholesterol targets.

Table 2 shows the multivariable associations between patient characteristics and statin use (among all patients) and high-intensity statin use (among statin users) at 1 year. The association between age and statin use was nonlinear, with patients aged 65 to 74 years having the highest odds of receiving statins. A prior history of CAD and CeVD (ie, other ASCVD), diabetes, hypertension, smoking, obesity (BMI ≥ 30 kg/m²), and severe PAD were strongly associated with higher odds of statin use. In contrast, female sex and Black race were associated with lower odds of receiving statins. However, among patients already on a statin, older age was associated with progressively lower odds of receiving high-intensity statins (Table 2). Black race was associated with higher odds of high-intensity statin use (among statin users). The associations between preexisting ASCVD, smoking status, obesity (BMI ≥ 30 kg/m²), and PAD severity with high-intensity statin use were similar to their associations

with overall statin use. The rate of overall and high-intensity statin use was modestly higher in patients who underwent lower extremity revascularization within 6 months of PAD diagnosis.

At the facility level, the median statin use rate was 70% (interquartile range, 67%–73%), ranging from 57% to 78% across facilities (Figure 3A). Facility-level variation in high-intensity statin use was greater, with a median rate of 36% (interquartile range, 33%–43%; range, 20%–54%) (Figure 3B). The median facility random-effect odds ratio for any statin use was 1.18 (95% CI, 1.14–1.21), and for high-intensity statin use was 1.39 (95% CI, 1.32–1.46). These estimates indicate that, for 2 identical patients with a new diagnosis of PAD treated at 2 randomly selected VA sites, the odds of receiving any statin differ by 18% and, among statin users, the odds of receiving high-intensity statins differ by 39%. Using the facility predicted-to-expected ratios and their corresponding CIs, we identified 11 facilities with significantly higher and 14 with

Table 2. Characteristics Associated With Any Statin Use Versus No Use and High-Intensity Use Versus Low/Moderate-Intensity Use, Using Multivariable Logistic Regression

Characteristic	Any intensity vs no statin use, OR (95% CI)	High intensity vs low/moderate intensity, OR (95% CI)
Age, y (reference: <55)		
55–64	1.30 (1.19–1.43)	0.87 (0.77–0.99)
65–74	1.64 (1.50–1.80)	0.70 (0.62–0.79)
≥75	1.21 (1.10–1.33)	0.42 (0.37–0.48)
Sex (reference: male)		
Female	0.67 (0.61–0.73)	0.94 (0.84–1.05)
Race (reference: White)		
Black	0.84 (0.81–0.88)	1.15 (1.09–1.20)
Other	0.86 (0.81–0.92)	1.05 (0.97–1.13)
Ethnicity (reference: not Hispanic/Latino)		
Hispanic or Latino	1.08 (0.98–1.18)	1.15 (1.04–1.27)
Unknown	1.04 (0.95–1.14)	1.00 (0.90–1.10)
Prior ASCVD (reference: no prior ASCVD)		
Prior CAD only	2.07 (1.99–2.15)	2.54 (2.44–2.65)
Prior CeVD only	1.58 (1.50–1.65)	1.52 (1.44–1.61)
Prior CAD and CeVD	2.49 (2.38–2.61)	3.20 (3.05–3.37)
Comorbidities		
Diabetes	1.56 (1.52–1.61)	1.19 (1.15–1.24)
Hypertension	1.71 (1.63–1.80)	1.19 (1.11–1.27)
Chronic kidney disease	1.03 (0.99–1.06)	1.02 (0.98–1.06)
COPD	1.01 (0.97–1.04)	0.96 (0.93–0.99)
Heart failure	1.01 (0.97–1.05)	1.28 (1.23–1.33)
Cancer	0.94 (0.91–0.98)	0.97 (0.93–1.02)
Dementia	0.67 (0.63–0.71)	0.88 (0.81–0.94)
Depression	0.95 (0.92–0.98)	1.07 (1.04–1.11)
Liver disease	0.69 (0.66–0.72)	0.83 (0.78–0.87)
Smoking status (reference: never)		
Current	1.01 (0.96–1.06)	1.09 (1.03–1.15)
Former	1.22 (1.16–1.27)	1.06 (1.01–1.12)
Unknown	1.10 (1.04–1.15)	1.08 (1.02–1.14)
PAD severity (reference: mild)		
Moderate	1.07 (1.04–1.11)	1.11 (1.07–1.16)
Severe	1.07 (1.03–1.12)	1.16 (1.10–1.22)
Revascularization within 6 mo (reference: no)	1.12 (1.07–1.18)	1.26 (1.19–1.33)
BMI (kg/m ² ; reference: mild)		
≥30	1.21 (1.17–1.25)	1.06 (1.03–1.10)
Unknown	0.86 (0.81–0.91)	0.92 (0.86–0.99)
Facility random effect	1.18 (1.14–1.21)	1.39 (1.32–1.46)

ASCVD indicates atherosclerotic cardiovascular disease; BMI, body mass index; CAD, coronary artery disease; CeVD, cerebrovascular disease; COPD, chronic obstructive pulmonary disease; OR, odds ratio; and PAD, peripheral artery disease.

significantly lower overall statin utilization rates than the average. Similarly, among statin users, 22 facilities had significantly higher rates, and 20 had significantly

lower rates than the average high-intensity statin utilization rate.

DISCUSSION

In a national cohort of Veterans with new-onset PAD, we found that ≈70% of patients were on a statin within 1 year of PAD diagnosis. However, use of high-intensity statins was low at 37%, LDL-cholesterol was above goal (≤70 mg/dL) in >60% of patients, and <2% were receiving nonstatin lipid-lowering therapy. Age, race, and comorbidities, notably concomitant ASCVDs, were associated with statin and high-intensity statin use. After adjusting for differences in patient characteristics, there was modest variation in statin use across sites, especially in high-intensity statin use. Several of our findings warrant further discussion.

Using 1999 to 2004 National Health and Nutrition Examination Survey data, Pande et al¹⁴ reported that only 30.5% of patients with PAD were on a statin. More recent studies have reported statin use in PAD between 46.7% and 63%.^{7,10,13} In contrast, the overall use of statins of 69.8% within a year of PAD diagnosis in our study is higher than these prior reports. Some of the differences may be due to differences in the study population. In our study, PAD identification was based on ABI and TBI values, yielding much higher overall accuracy than studies relying on *International Classification of Diseases* diagnosis codes.^{15–19} Thus, it is possible that inclusion of patients without PAD, who lacked a clear indication for statin therapy, may partly explain the lower overall rates of statin use reported in prior studies. In addition, higher rates of statin use in our study may also be reflective of care in the VA health system, which has made a concerted effort to optimize risk-factor control in patients at high-risk of cardiovascular events.

Despite the higher overall use of statins, we also found some gaps in care quality. First, although our study included predominantly White men, the odds of active statin use were lower in women and in those who self-identified as Black. This is consistent with prior studies.^{29–33} Second, although use of statins was high, only about one-third of patients were on a high-intensity statin at 1-year from their PAD diagnosis, and 62% of patients were not at goal LDL-cholesterol. Moreover, a diagnosis of PAD is considered a risk enhancer in current guidelines, yet statin therapy was newly initiated or intensified in only a small proportion of patients after PAD diagnosis, and few were treated with nonstatin alternatives. This is despite strong evidence from multiple studies that have highlighted a benefit of high-intensity statins in reducing the risk of myocardial infarction and stroke, as well as progression to chronic limb-threatening ischemia and major amputation in patients with PAD.^{3–7,9,10} Barriers to optimal statin therapy have been previously reported and include patient (prior intolerance, poor adherence,

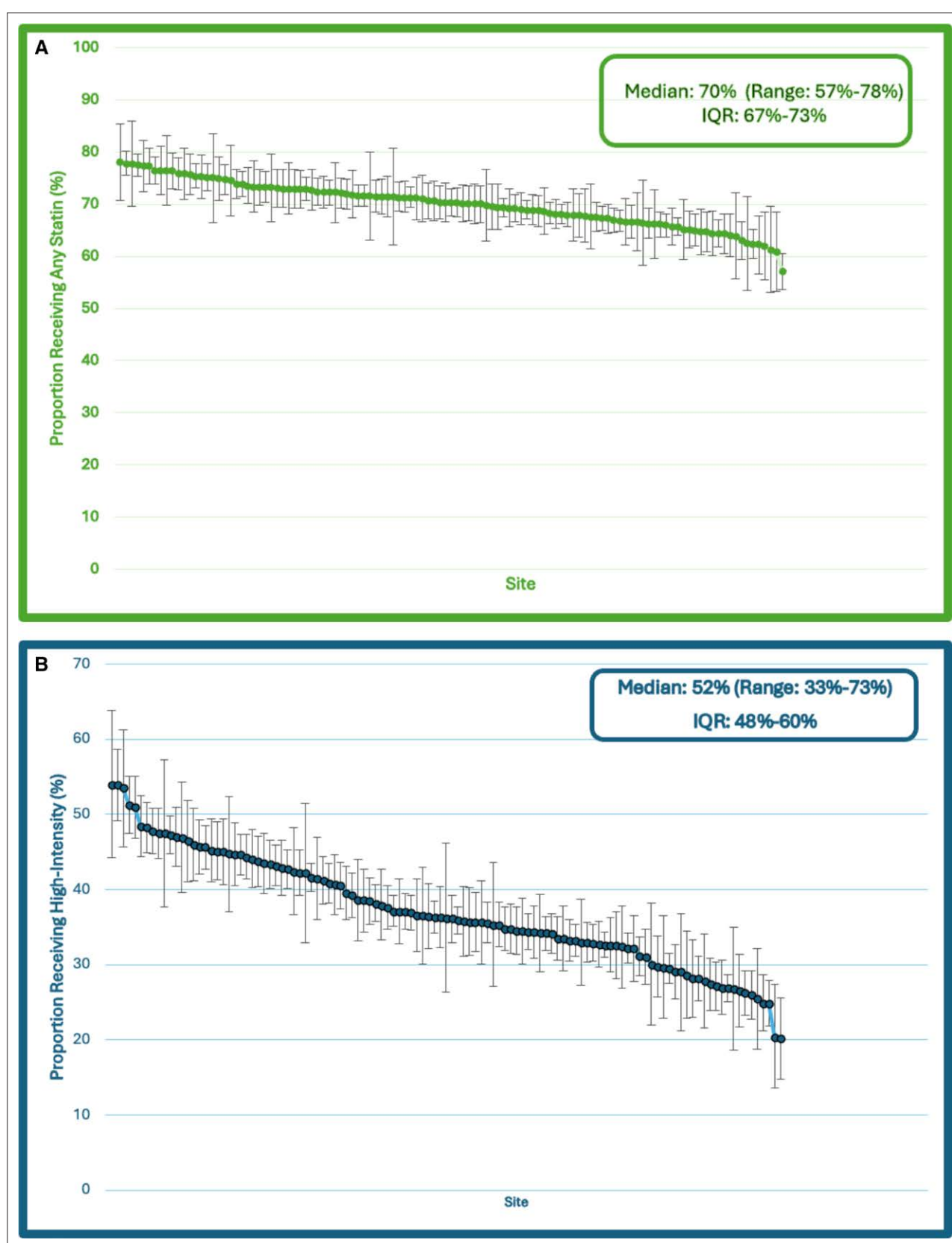


Figure 3. Site-level variation in the rates of any statin therapy (A) and high-intensity statin therapy (B) utilization 1 year after index peripheral artery disease (PAD) diagnosis.

Among 115 sites, there was modest variation across sites in the use of any statin (A), with a median odds ratio of 1.18 (95% CI, 1.14–1.21). Site-level variation in the use of high-intensity statins (B) was larger, with a median odds ratio of 1.39 (95% CI, 1.32–1.46). IQR indicates interquartile range.

misinformation) and clinician (therapeutic inertia) factors.³⁴ Prior real-world studies report a higher rate of statin intolerance than clinical trials where the treatment is blinded (nocebo effect).^{35,36} Similarly, in our cohort, the overall rate of documented statin intolerance was high (13.7%). While statin intolerance rates were higher in the nonstatin group compared with statin users, exclusion of patients with statin intolerance minimally changed the rates of overall and high-intensity statin use. Importantly, previous studies also show that patients who experience intolerance or allergic reactions to 1 statin often tolerate an alternative statin.^{37,38} This was also seen in our cohort, where >10% of patients actively taking statins had a previously documented allergy. Taken together, prior adverse effects and intolerance are unlikely to fully explain the suboptimal use of high-intensity statins. This is supported by the observation that more than 50% of statin nonusers at 1 year had not filled a statin prescription in the preceding 3 years, and fewer than 2% received nonstatin lipid-lowering therapies such as ezetimibe or PCSK-9 inhibitors. It is also possible that patients with PAD have a high burden of comorbidities and, therefore, may have multiple competing medical care needs, which can distract from optimizing secondary prevention therapies during clinical encounters.

We also found that statin use was lower in patients with PAD without concomitant ASCVD, likely due to the greater clinical attention to management in patients with polyvascular disease.¹³ Previous studies have also shown higher statin prescription rates following a new diagnosis of other ASCVDs (CAD or CeVD). For instance, Zheutlin and colleagues³⁹ showed that following an acute myocardial infarction or coronary revascularization, lipid-lowering therapy was intensified in 47% of patients (over 80% among those not previously on a statin). Similarly, Colantino et al¹³ found low statin prescription rates in patients with PAD as compared with those with other single-system ASCVDs or polyvascular disease. In our study, we found that <1 in 4 PAD-only patients were on high-intensity statins, while ≈3 in 4 remained above the LDL target. This gap persists despite evidence that PAD carries a comparable risk of future ASCVD events, as well as substantial morbidity and mortality when compared with CAD or CeVD. Potential explanations include patient reluctance to initiate statins, which is not uncommon, provider inertia, a higher perceived acuity of CAD and CeVD relative to PAD, limited patient awareness of PAD-associated risks, and variability in specialists' involvement in PAD care, including cardiology, primary care, vascular surgery, and geriatrics. Targeted efforts to improve statin initiation and intensification in PAD, particularly through multispecialty collaboration, are essential to closing these gaps.

Similarly, we found a modest positive association between lower-extremity revascularization within 6 months and overall statin and high-intensity statin use.

This finding is consistent with prior studies that showed slight improvements in statin optimization anchored around revascularization procedures.^{40,41} Several factors could explain these findings: patients undergoing revascularization may be more likely to receive optimized medical management, including guideline-directed statin therapy. Additionally, patients undergoing revascularization often have more frequent interactions with the healthcare system, creating more opportunities for statin initiation or intensification.

We also found modest site-level variation in the use of high-intensity statins even after adjusting for patient variables. Although the median rate was 37%, use of high-intensity statins in PAD approached 54% at some sites. Close to one-fifth of the VA sites in our study (22/115) were statistically better performers in high-intensity statin utilization. These findings highlight the presence of higher-performing facilities within the VA system, particularly in high-intensity statin prescription rates. It is possible that these VA sites have achieved higher-than-average rates of high-intensity statin use through the implementation of care innovations. Identifying and systematically studying these sites may provide valuable insights into effective implementation strategies, local workflows, and system-level practices that could be leveraged to improve statin utilization more broadly across the network. In a recent randomized controlled trial in the VA, individualized reminders to clinicians before their visit led to an absolute 10% increase in high-intensity statin prescribing among patients with ASCVD, including PAD.⁴² A separate study showed that a telemedicine intervention using an algorithmic approach, led primarily by nonclinician navigators with clinical pharmacist support, was highly effective in statin initiation, intensification, use of nonstatin lipid-lowering therapy, and cholesterol level optimization.⁴³ Wide dissemination of such strategies to ensure that patients at high risk of clinical events are appropriately treated is warranted. Outside of randomized controlled trials, future studies can identify and study high-performing centers to better understand existing best practices and implementation strategies adopted at sites with higher statin use. Such efforts could be studied using a mixed-methods approach that integrates quantitative analyses with qualitative assessments (eg, provider workflows, institutional culture, and patient-level barriers) to inform scalable interventions with targetable objective end points.

Our study findings need to be interpreted in the context of the following limitations. First, we used VA pharmacy and Medicare Part D claims to evaluate medication use. Patients filling their medications using self-pay or private insurance would be missed, leading to a potential underestimation of statin use. Second, we estimated statin use based on pharmacy dispensing information, which may not represent actual patient use. Third, we lacked information to differentiate whether nonuse or use of

low/moderate-intensity statin was due to limited access, a prior history of adverse effects (beyond what was documented as an allergic reaction), or therapeutic inertia. Since VA pharmacy benefits are generous and most patients have limited or no copayments, financial barriers are unlikely to be a major factor. Moreover, although we report documented allergic reactions and adverse events attributed to statins, definitions of statin intolerance for this analysis are not standardized and are based on clinician documentation. Variability in documentation practices from site to site and provider to provider may lead to both under-recognition of true statin intolerance and over-attribution of symptoms unrelated to statin therapy. Fourth, women and racial/ethnic minorities were underrepresented in our cohort, limiting generalizability. Moreover, our findings reflect care in the VA, and their generalizability to non-VA health systems may be limited. Fifth, it is possible that patients not on a statin had been informed of its benefits but still refused treatment. Finally, while we highlight patterns of statin use in our cohort, future studies should consider examining the association between statin adherence and long-term outcomes in patients with PAD.

In conclusion, although $\approx 70\%$ of patients with PAD were receiving a statin within 1 year of PAD diagnosis, use of high-intensity statins and nonstatin lipid-lowering therapy remained low, and a substantial proportion of patients were not at goal LDL-cholesterol. Given the considerable benefit of high-intensity statins in reducing the risk of clinical events in PAD, strategies focused on optimizing lipid-lowering therapy in PAD are warranted.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Tables S1–S7

REFERENCES

- Arnett DK, Blumenthal RS, Albert MA, Buroker AB, Goldberger ZD, Hahn EJ, Himmelfarb CD, Khera A, Lloyd-Jones D, McEvoy JW, et al. 2019 ACC/AHA guideline on the primary prevention of cardiovascular disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation*. 2019;140:e596–e646. doi: 10.1161/CIR.0000000000000678
- Ramos R, Garcia-Gil M, Comas-Cufi M, Quesada M, Marrugat J, Elosua R, Sala J, Grau M, Marti R, Ponjoan A, et al. Statins for prevention of cardiovascular events in a low-risk population with low ankle brachial index. *J Am Coll Cardiol*. 2016;67:630–640. doi: 10.1016/j.jacc.2015.11.052
- Newman AB, Siscovick DS, Manolio TA, Polak J, Fried LP, Borhani NO, Wolfson SK. Ankle-arm index as a marker of atherosclerosis in the Cardiovascular Health Study. Cardiovascular Health Study (CHS) Collaborative Research Group. *Circulation*. 1993;88:837–845. doi: 10.1161/01.cir.88.3.837
- Resnick HE, Lindsay RS, McDermott MM, Devereux RB, Jones KL, Fabsitz RR, Howard BV. Relationship of high and low ankle brachial index to all-cause and cardiovascular disease mortality: the Strong Heart Study. *Circulation*. 2004;109:733–739. doi: 10.1161/01.CIR.0000112642.63927.54
- Vogt MT, Cauley JA, Newman AB, Kuller LH, Hulley SB. Decreased ankle/arm blood pressure index and mortality in elderly women. *JAMA*. 1993;270:465–469
- Willey J, Mentias A, Vaughan-Sarrazin M, McCoy K, Rosenthal G, Girotra S. Epidemiology of lower extremity peripheral artery disease in veterans. *J Vasc Surg*. 2018;68:527–535.e5. doi: 10.1016/j.jvs.2017.11.083
- Arya S, Khakharia A, Binney ZO, DeMartino RR, Brewster LP, Goodney PP, Wilson PWF. Association of statin dose with amputation and survival in patients with peripheral artery disease. *Circulation*. 2018;137:1435–1446. doi: 10.1161/CIRCULATIONAHA.117.032361
- Foley TR, Singh GD, Kokkinidis DG, Choy HK, Pham T, Amsterdam EA, Rutledge JC, Waldo SW, Armstrong EJ, Laird JR. High-intensity statin therapy is associated with improved survival in patients with peripheral artery disease. *J Am Heart Assoc*. 2017;6:e005699. doi: 10.1161/JAHA.117.005699
- Heart Protection Study Collaborative G. Randomized trial of the effects of cholesterol-lowering with simvastatin on peripheral vascular and other major vascular outcomes in 20,536 people with peripheral arterial disease and other high-risk conditions. *J Vasc Surg*. 2007;45:645–654; discussion 653–644. doi: 10.1016/j.jvs.2006.12.054
- Kumbhani DJ, Steg PG, Cannon CP, Eagle KA, Smith SC Jr, Goto S, Ohman EM, Elbez Y, Sritara P, Baumgartner I, et al; REACH Registry Investigators. Statin therapy and long-term adverse limb outcomes in patients with peripheral artery disease: insights from the REACH registry. *Eur Heart J*. 2014;35:2864–2872. doi: 10.1093/eurheartj/ehu080
- Gerhard-Herman MD, Gornik HL, Barrett C, Barshe NR, Corriere MA, Drachman DE, Fleisher LA, Fowkes FG, Hamburg NM, Kinlay S, et al. 2016 AHA/ACC guideline on the management of patients with lower extremity peripheral artery disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Circulation*. 2017;135:e686–e725. doi: 10.1161/CIR.00000000000000470
- Gornik HL, Aronow HD, Goodney PP, Arya S, Brewster LP, Byrd L, Chandra V, Drachman DE, Eaves JM, Ehrman JK, et al; Peer Review Committee Members. 2024 ACC/AHA/AACVPR/APMA/ABC/SCAI/SVM/SVN/SVS/SIR/VES guideline for the management of lower extremity peripheral artery disease: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2024;149:e1313–e1410. doi: 10.1161/CIR.00000000000001251
- Colantonio LD, Hubbard D, Monda KL, Mues KE, Huang L, Dai Y, Jackson EA, Brown TM, Rosenson RS, Woodward M, et al. Atherosclerotic risk and statin use among patients with peripheral artery disease. *J Am Coll Cardiol*. 2020;76:251–264. doi: 10.1016/j.jacc.2020.05.048
- Pande RL, Perlstein TS, Beckman JA, Creager MA. Secondary prevention and mortality in peripheral artery disease: National Health and Nutrition Examination Study, 1999 to 2004. *Circulation*. 2011;124:17–23. doi: 10.1161/CIRCULATIONAHA.110.003954
- Fan J, Arruda-Olson AM, Leibson CL, Smith C, Liu G, Bailey KR, Kullo IJ. Billing code algorithms to identify cases of peripheral artery disease from administrative data. *J Am Med Inform Assoc*. 2013;20:e349–e354. doi: 10.1136/amiainl-2013-001827
- Hong Y, Sebastianski M, Makowsky M, Tsuyuki R, McMurtry MS. Administrative data are not sensitive for the detection of peripheral artery disease in the community. *Vasc Med*. 2016;21:331–336. doi: 10.1177/1358863X16631041
- Lasota AN, Overvad K, Eriksen HH, Tjønneland A, Schmidt EB, Gronholdt MM. Validity of peripheral arterial disease diagnoses in the

- Danish National Patient Registry. *Eur J Vasc Endovasc Surg*. 2017;53:679–685. doi: 10.1016/j.ejvs.2016.12.031
18. Weissler EH, Zhang J, Lippmann S, Rusincovitch S, Henao R, Jones WS. Use of natural language processing to improve identification of patients with peripheral artery disease. *Circ Cardiovasc Interv*. 2020;13:e009447. doi: 10.1161/CIRCINTERVENTIONS.120.009447
 19. Friberg JE, Qazi AH, Boyle B, Franciscus C, Vaughan-Sarrazin M, Westerman D, Patterson OV, Parr SK, Matheny ME, Arya S, et al. Ankle- and toe-brachial index for peripheral artery disease identification: unlocking clinical data through novel methods. *Circ Cardiovasc Interv*. 2022;15:e011092. doi: 10.1161/CIRCINTERVENTIONS.121.011092
 20. Rasmussen P, Farmer CM. The promise and challenges of VA community care: veterans' issues in focus. *Rand Health Q*. 2023;10:9
 21. Wagner TH, Schmidt A, Belli F, Aouad M, Gehlert E, Desai M, Graham L, Rose L. Health insurance enrollment among US veterans, 2010–2021. *JAMA Netw Open*. 2024;7:e2430205. doi: 10.1001/jamanetworkopen.2024.30205
 22. VA DATAHUB. Accessed March 29. <https://www.datahub.va.gov/stories/s/Use-of-VA-Benefits-and-Services-2023-Part-1-/59yt-v847/>
 23. Girotra S, Li Q, Vaughan-Sarrazin M, Lund BC, Al-Garadi M, Beckman JA, Nathani R, Hoffman RM, Chan PS, Banerjee S, et al. Long-term outcomes of peripheral artery disease in veterans: analysis of the Peripheral Artery Disease Long-Term Survival study (PEARLS). *J Am Heart Assoc*. 2025;14:e038403. doi: 10.1161/JAHA.124.038403
 24. Quan H, Sundararajan V, Halfon P, Fong A, Burnand B, Luthi JC, Saunders LD, Beck CA, Feasby TE, Ghali WA. Coding algorithms for defining comorbidities in ICD-9-CM and ICD-10 administrative data. *Med Care*. 2005;43:1130–1139. doi: 10.1097/01.mlr.0000182534.19832.83
 25. Elixhauser A, Steiner C, Harris DR, Coffey RM. Comorbidity measures for use with administrative data. *Med Care*. 1998;36:8–27. doi: 10.1097/00005650-199801000-00004
 26. Goldstein H, Browne W, Rasbash J. Multilevel modelling of medical data. *Stat Med*. 2002;21:3291–3315. doi: 10.1002/sim.1264
 27. Goldstein H, William B, Rasbash J. Partitioning variation in multilevel models. *Underst Stat*. 2002;1:223–231. doi: 10.1207/S15328031US0104_02
 28. Larsen K, Merlo J. Appropriate assessment of neighborhood effects on individual health: integrating random and fixed effects in multilevel logistic regression. *Am J Epidemiol*. 2005;161:81–88. doi: 10.1093/aje/kwi017
 29. Hirsch AT, Allison MA, Gomes AS, Corriere MA, Duval S, Ershow AG, Hiatt WR, Karas RH, Lovell MB, McDermott MM, et al; American Heart Association Council on Peripheral Vascular Disease. A call to action: women and peripheral artery disease: a scientific statement from the American Heart Association. *Circulation*. 2012;125:1449–1472. doi: 10.1161/CIR.0b013e31824c39ba
 30. Allison MA, Armstrong DG, Goodney PP, Hamburg NM, Kirksey L, Lancaster KJ, Mena-Hurtado CI, Misra S, Treat-Jacobson DJ, White Solaru KT; American Heart Association Council on Peripheral Vascular Disease; Council on Hypertension; and Council on Lifestyle and Cardiometabolic Health. Health disparities in peripheral artery disease: a scientific statement from the American Heart Association. *Circulation*. 2023;148:286–296. doi: 10.1161/CIR.0000000000001153
 31. Durazzo TS, Frencher S, Gusberg R. Influence of race on the management of lower extremity ischemia: revascularization vs amputation. *JAMA Surg*. 2013;148:617–623. doi: 10.1001/jamasurg.2013.1436
 32. Krawisz AK, Natesan S, Wadhwa RK, Chen S, Song Y, Yeh RW, Jaff MR, Giri J, Julien H, Secemsky EA. Differences in comorbidities explain black-white disparities in outcomes after femoropopliteal endovascular intervention. *Circulation*. 2022;146:191–200. doi: 10.1161/CIRCULATIONAHA.122.058998
 33. Hackler EL, Hamburg NM, White Solaru KT. Racial and ethnic disparities in peripheral artery disease. *Circ Res*. 2021;128:1913–1926. doi: 10.1161/CIRCRESAHA.121.318243
 34. Virani SS, Ballantyne CM, Petersen LA. Guideline-concordant statin therapy use in secondary prevention: Should the medical community wait for divine intervention? *J Am Coll Cardiol*. 2022;79:1814–1817. doi: 10.1016/j.jacc.2022.02.042
 35. Howard JP, Wood FA, Finegold JA, Nowbar AN, Thompson DM, Arnold AD, Rajkumar CA, Connolly S, Cegla J, Stride C, et al. Side effect patterns in a crossover trial of statin, placebo, and no treatment. *J Am Coll Cardiol*. 2021;78:1210–1222. doi: 10.1016/j.jacc.2021.07.022
 36. Herrett E, Williamson E, Brack K, Beaumont D, Perkins A, Thayne A, Shakur-Still H, Roberts I, Prowse D, Goldacre B, et al; StatinWISE Trial Group. Statin treatment and muscle symptoms: series of randomised, placebo controlled n-of-1 trials. *BMJ*. 2021;372:n135. doi: 10.1136/bmj.n135
 37. Cheeley MK, Saseen JJ, Agarwala A, Ravilla S, Ciffone N, Jacobson TA, Dixon DL, Maki KC. NLA scientific statement on statin intolerance: a new definition and key considerations for ASCVD risk reduction in the statin intolerant patient. *J Clin Lipidol*. 2022;16:361–375. doi: 10.1016/j.jacl.2022.05.068
 38. Ruscica M, Bertolotti A, Gobbi C, Sirtori CR, Carugo S, Corsini A. Lipid-lowering approaches to manage statin-intolerant patients. *Eur Heart J Suppl*. 2024;26:i56–i59. doi: 10.1093/eurheartjsupp/suae007
 39. Zheutlin AR, Derington CG, Herrick JS, Rosenson RS, Poudel B, Safford MM, Brown TM, Jackson EA, Woodward M, Reading S, et al. Lipid-lowering therapy use and intensification among United States veterans following myocardial infarction or coronary revascularization between 2015 and 2019. *Circ Cardiovasc Qual Outcomes*. 2022;15:e008861. doi: 10.1161/CIRCOUTCOMES.121.008861
 40. Singh N, Ding L, Devera J, Magee GA, Garg PK. Prescribing of statins after lower extremity revascularization procedures in the US. *JAMA Netw Open*. 2021;4:e2136014. doi: 10.1001/jamanetworkopen.2021.36014
 41. Smolderen KG, Romain G, Provance JB, Scierka LE, Mao J, Goodney PP, Henke PK, Sedrakyan A, Mena-Hurtado C. Guideline-directed medical therapy and long-term mortality and amputation outcomes in patients undergoing peripheral vascular interventions. *JACC Cardiovasc Interv*. 2023;16:332–343. doi: 10.1016/j.jcin.2022.09.022
 42. Virani SS, Ramsey DJ, Westerman D, Kuebler MK, Chen L, Akeroyd JM, Gobbel GT, Ballantyne CM, Petersen LA, Turchin A, et al. Cluster randomized trial of a personalized clinical decision support intervention to improve statin prescribing in patients with atherosclerotic cardiovascular disease. *Circulation*. 2023;147:1411–1413. doi: 10.1161/CIRCULATIONAHA.123.064226
 43. Blood AJ, Cannon CP, Gordon WJ, Mailly C, MacLean T, Subramaniam S, Tucci M, Crossen J, Nichols H, Waghlikar KB, et al. Results of a remotely delivered hypertension and lipid program in more than 10 000 patients across a diverse health care network. *JAMA Cardiol*. 2023;8:12–21. doi: 10.1001/jamacardio.2022.4018